chapter 24

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ElectroMagnetism

Magnetic field due to Long straight Wire:

The magnitude of the magnetic field at a perpendicular distance R from a long (infinite) straight wire carrying a current i given by:

$$B = \frac{\mu_0 i}{2\pi R} \quad \text{(long straight wire)}$$

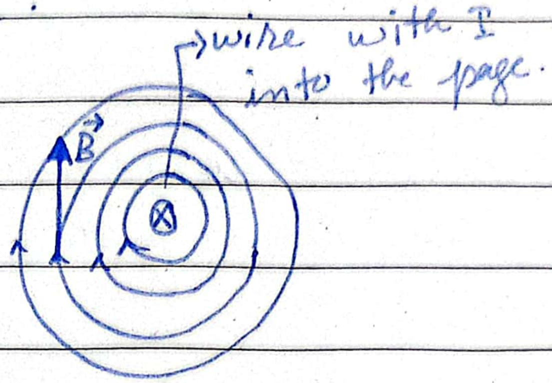
Permeability:

In electromagnetism, Permeability is the measure of the ability of a material to support the formation of a magnetic field within itself.

It is the degree of magnetization that a material obtains in the response to an applied magnetic field.

$$\mu_0 = 4\pi \times 10^{-7}\ Tm/A \approx 1.26 \times 10^{-6}\ Tm/A$$

The magnetic field vector at any point is tangent to a circle.



By right hand rule, the thumb is in the current's direction. The fingers reveal the field vector's direction, which is tangent to a circle.

Ampere's Law:

The line integral $\oint \vec{B}\cdot d\vec{s}$ of the magnetic field $\vec{B}$ along any closed path is equal to the total current enclosed inside the path multiplied by μ_0.

$$\oint \vec{B}\cdot d\vec{s} = \mu_0 i_{enc}$$

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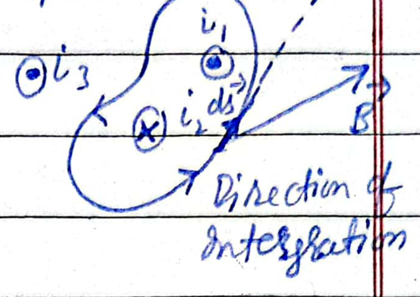
The $d\vec{s}$ is the current-length element vector along the direction of an Amperian loop.

Only the currents encircled by the loop are used in Ampere's law.



Determination of all B·ds. The closed path is divided into n elements and take their sums.

$$\oint \vec{B}\cdot d\vec{s} = \mu_0 i_{enc}$$

- Curl your ~~(fingers)~~ right hand around the Amperian loop, with the fingers pointing in the direction of integration.

→ All currents inside the loop parallel to thumb are counted as positive, while anti-parallel counted as negative.

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→ All currents outside the loop are not counted.

Ampere's Law, $\vec{B}$ outside a Long current carrying wire:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}$$

$$\oint \vec{B} \cdot d\vec{s} = \oint B\cos\theta \, ds = B \oint ds$$
$$= B(2\pi r)$$

$$B(2\pi r) = \mu_0 i$$

$$B = \frac{\mu_0 i}{2\pi r}$$ (outside of straight wire)

Ampere's Law, $\vec{B}$ inside a Long current carrying wire:

$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}$

$$\oint \vec{B} \cdot d\vec{s} = B \oint ds = B(2\pi r)$$

$\oint$ → surface integral

$\oint$ $i_{enc} = ?$

$$\vec{J}_{total} = \vec{J}$$

$$\frac{\text{total current } (i)}{\text{total Area of } R} = \frac{\text{Current enclosed } (i_{enc})}{\text{Area enclosed by } r}$$

area of circle.

$$\frac{i}{\pi R^2} = \frac{i_{enc.}}{\pi r^2}$$

$$i_{enc.} = \frac{i\, r^2}{R^2}$$

Now,

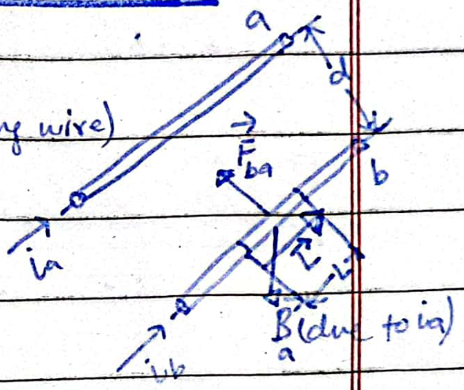
$$\oint \vec{B}\cdot d\vec{s} = \mu_0 i_{enc.}$$

$$B(2\pi r) = \mu_0 i \frac{r^2}{R^2}$$

$$B = \left(\frac{\mu_0 i}{2\pi R^2}\right) r$$ (inside straight wire)

Force Between Two Parallel wires:

$B_a = \dfrac{\mu_0 i_a}{2\pi d}$ $\because B = \dfrac{\mu_0 i}{2\pi R}$ (long wire)



$$\vec{F}_{ba} = i_b \vec{L} \times \vec{B}_a$$

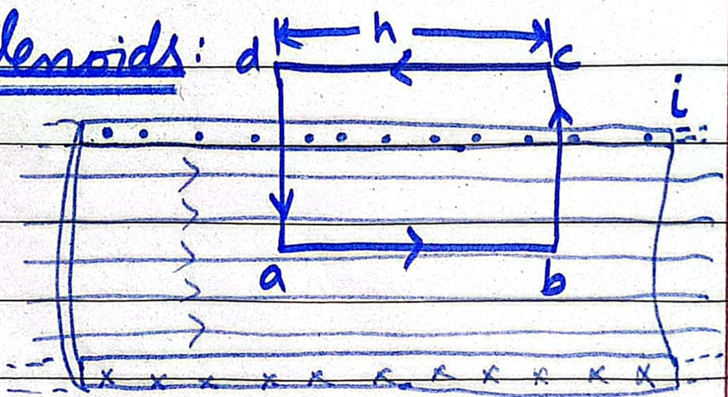
$$\vec{F}_{ba} = i_b L B_a \sin 90^\circ = \frac{\mu_0 L i_a i_b}{2\pi d}$$

The field due to (a) at the position of (b) creates a force on (b).

The Solenoid:

The solenoid is a long, tightly wound helical wire coil in which the coil length is much larger than the coil diameter. Viewing the solenoid as a collection of single circular loops, one can see that the magnetic field inside is approximately uniform.

Solenoids:



$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc} \quad \text{(Ampere's law)}$$

We will use the Amperian loop abcd. It is a rectangle with its long side parallel

to the (side) solenoid axis.
One long side (ab) is inside the solenoid, while the other (cd) is outside:

$$\oint \vec{B}\cdot d\vec{s} = \int_a^b \vec{B}\cdot d\vec{s} + \int_b^c \vec{B}\cdot d\vec{s} + \int_c^d \vec{B}\cdot d\vec{s} + \int_d^a \vec{B}\cdot d\vec{s}$$

$$\int_a^b \vec{B}\cdot d\vec{s} = \int_a^b B d \cos 0 = B\int_a^b ds = Bh$$

$$\int_b^c \vec{B} d\vec{s} = \int_c^d \vec{B} d\vec{s} = \int_d^a \vec{B}\cdot d\vec{s} = 0$$

So.

$$\oint \vec{B}\cdot d\vec{s} = Bh$$

$$\oint \vec{B}\cdot d\vec{s} = \mu_0 i_{enc}$$

$$Bh = \mu_0 n h i$$

$$\boxed{B = \mu_0 n i}$$

The enclosed current

$i_{enc.} = nhi$

n = numbers of turns per unit length.

$n = N/h$ h = length of solenoid

N = number of turns.

$i_{enc.} = (i\, N/h)\, h = Ni$

n = no. of turns per unit length of solenoid.

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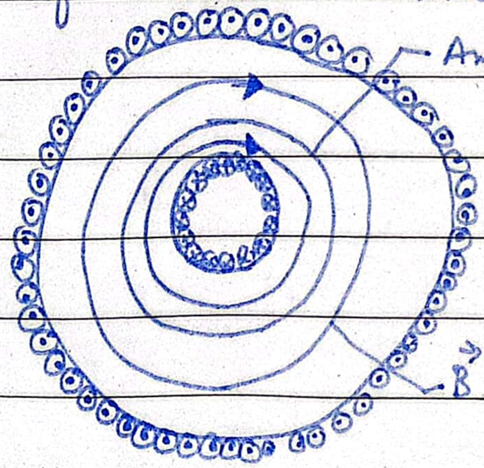
Magnetic field of Toroid:

A Toroid which we may describe as a (hollow) Solenoid that has been curved untill its two ends meet, forming a sort of hollow bracelet.



current enlcosed $i_{enc.} = Ni$

$$B(2\pi r) = \mu_0 i N,$$

$$B = \frac{\mu_0 N i}{2\pi r} \quad (\text{toroid})$$